

C Programming Language

Comprehensive Theory, Core Concepts, Code Examples & 25 Practice Questions

1. Introduction to C Programming

C is a powerful procedural, general-purpose programming language developed by Dennis Ritchie at Bell Labs in 1972. It is considered the mother of modern programming languages and serves as the foundation for operating systems (like Linux and Windows), embedded systems, and compilers.

Key Features of C Language

- **Procedural Language:** Instructions are executed step-by-step in a structured approach using functions.
- **Fast and Efficient:** Compiles directly to machine code, offering low-level memory access.
- **Portability:** C programs can run on different platforms with little to no modification.
- **Rich Set of Operators:** Includes arithmetic, relational, logical, bitwise, and assignment operators.
- **Modularity:** Code can be divided into smaller reusable units called functions.

2. Core C Concepts & Code Examples

1. Structure of a C Program & Variables

Every C program starts execution from the `main()` function. Header files like `<stdio.h>` are included for input/output operations. Variables must be declared with explicit data types (`int`, `float`, `char`, `double`).

```
#include <stdio.h>

int main() {
    int age = 20;
    float salary = 45500.50;
    char grade = 'A';

    printf("Age: %d, Grade: %c, Salary: %.2f\n", age, grade, salary);
    return 0;
}
```

2. Decision Making & Loops

Conditional logic is handled via `if-else` and `switch-case` statements. Loops (`for`, `while`, `do-while`) iterate over code blocks until a condition becomes false.

```
#include <stdio.h>

int main() {
    int num = 10;
    if (num % 2 == 0) {
        printf("Even\n");
    } else {
        printf("Odd\n");
    }

    for (int i = 1; i <= 3; i++) {
        printf("Count: %d\n", i);
    }
    return 0;
}
```

3. Arrays & Strings

An array is a fixed-size collection of elements of the same data type stored in contiguous memory locations. Strings in C are null-terminated (`'\0'`) character arrays.

```
#include <stdio.h>

int main() {
    int marks[3] = {85, 90, 95};
    char name[] = "C Programming";

    printf("First Mark: %d\n", marks[0]);
    printf("Language: %s\n", name);
    return 0;
}
```

4. Functions & Scope

Functions divide code into manageable blocks. Arguments can be passed by value (copying data) or by reference (passing memory addresses via pointers).

```
#include <stdio.h>

int add(int a, int b) {
    return a + b;
}

int main() {
    int sum = add(12, 8);
    printf("Sum = %d\n", sum);
    return 0;
}
```

5. Pointers & Memory Management

A pointer is a variable that stores the memory address of another variable. Pointers enable direct memory manipulation, dynamic memory allocation (malloc, calloc, free), and efficient array handling.

```
#include <stdio.h>

int main() {
    int val = 50;
    int *ptr = &val; // ptr stores address of val

    printf("Value: %d\n", val);
    printf("Address: %p\n", ptr);
    printf("Value via Pointer: %d\n", *ptr);
    return 0;
}
```

3. Structures, Unions & File I/O

- **Structures (struct):** User-defined data types that group variables of different types under a single name.
- **Unions (union):** Similar to structures, but all members share the exact same memory location.
- **File Handling:** Functions like fopen(), fprintf(), fscanf(), and fclose() manage file operations.
- **Preprocessor Directives:** Lines starting with # (e.g., #define MAX 100, #include) processed before compilation.

4. C Language Practice Questions (25 MCQs)

1. Who is known as the father of C language?

- | | |
|----------------------|-------------------|
| A) James Gosling | B) Dennis Ritchie |
| C) Bjarne Stroustrup | D) Ken Thompson |

2. C language was developed at which laboratory?

- | | |
|-------------------|----------------|
| A) AT&T Bell Labs | B) IBM Labs |
| C) Microsoft Labs | D) Google Labs |

3. Which function is mandatory in every C program?

- | | |
|------------|-------------|
| A) start() | B) main() |
| C) run() | D) printf() |

4. Which header file is required for printf() and scanf() functions?

- | | |
|---------------|---------------|
| A) <stdlib.h> | B) <stdio.h> |
| C) <conio.h> | D) <string.h> |

5. What is the correct format specifier for printing an integer in C?

- | | |
|-------|-------|
| A) %f | B) %d |
| C) %c | D) %s |

6. What is the character used to terminate every statement in C?

- | | |
|--------------|------------------|
| A) : (Colon) | B) ; (Semicolon) |
| C) . (Dot) | D) , (Comma) |

7. Which operator is used to get the address of a variable in C?

- | | |
|------|-------|
| A) * | B) & |
| C) % | D) -> |

8. What is the default return type of the main() function in ANSI C?

- | | |
|----------|---------|
| A) void | B) int |
| C) float | D) char |

9. Which operator is used for dereferencing a pointer variable?

- | | |
|------|-------|
| A) & | B) * |
| C) # | D) -> |

10. What character automatically terminates all C strings?

- | | |
|---------|---------|
| A) '\n' | B) '\0' |
| C) '\t' | D) '' |

11. What is the size of an int data type in standard 32-bit systems?

- | | |
|------------|------------|
| A) 2 Bytes | B) 4 Bytes |
| C) 8 Bytes | D) 1 Byte |

12. Which keyword is used to allocate memory dynamically in C?

- | | |
|----------|-----------|
| A) new | B) malloc |
| C) alloc | D) create |

13. Which loop is guaranteed to execute at least once?

- | | |
|---------------|------------------|
| A) for loop | B) do-while loop |
| C) while loop | D) nested loop |

14. What function is used to release dynamically allocated memory?

- | | |
|-------------|--------------|
| A) delete() | B) free() |
| C) remove() | D) release() |

15. What will be the output of integer division 7 / 2 in C?

- | | |
|--------|--------|
| A) 3.5 | B) 3 |
| C) 4 | D) 3.0 |

16. Which operator returns the remainder of an integer division?

- | | |
|--------|------|
| A) / | B) % |
| C) rem | D) # |

17. Which keyword is used to define user-defined custom data types?

- A) typedef
- B) struct
- C) type
- D) define

18. What is the first index position of an array in C?

- A) 1
- B) 0
- C) -1
- D) Depends on system

19. Which library function is used to calculate string length?

- A) strlen()
- B) strlen()
- C) len()
- D) stringlen()

20. What operator is used to access structure members using a structure pointer?

- A) . (Dot)
- B) -> (Arrow)
- C) * (Asterisk)
- D) & (Ampersand)

21. What does the sizeof operator return?

- A) Value of variable
- B) Size in bytes
- C) Address of variable
- D) Number of bits

22. Which preprocessor directive is used to include header files?

- A) #define
- B) #include
- C) #import
- D) #use

23. What mode opens a file for writing in C (overwriting existing content)?

- A) "r"
- B) "w"
- C) "a"
- D) "r+"

24. What happens when a function calls itself?

- A) Iteration
- B) Recursion
- C) Inline call
- D) Overloading

25. Which keyword is used to break out of a switch block?

- A) exit
- B) break
- C) continue
- D) return

Answer Key

- | | | | | |
|------|-------|-------|-------|-------|
| 1. B | 6. B | 11. B | 16. B | 21. B |
| 2. A | 7. B | 12. B | 17. A | 22. B |
| 3. B | 8. B | 13. B | 18. B | 23. B |
| 4. B | 9. B | 14. B | 19. B | 24. B |
| 5. B | 10. B | 15. B | 20. B | 25. B |